

Ahmed Attia

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Location: Masdar City, Abu Dhabi, UAE

EDUCATION

- **Mohamed bin Zayed University of Artificial Intelligence** Abu Dhabi, United Arab Emirates
Master of Science in Natural Language Processing; GPA: 3.7/4.0 Aug 2024 - Present
- **Kafr El-Sheikh University Faculty of Artificial Intelligence** Kafr El-Sheikh, Egypt
Bachelor of Science in Artificial Intelligence; GPA: 3.6/4.0 Aug 2020 - Jul 2024

EXPERIENCE

- **Student Resident at MBZUAI – Institute of Foundation Models** Abu Dhabi, UAE
Part-time Feb 2026 - May 2026
 - Led the reinforcement learning reasoning post-training of **Jais-3**, fine-tuning SFT checkpoints on Mathematical Reasoning, Code Generation using Hugging Face's **TRL**.
 - Helped curate the Arabic Math and Code datasets tailored to Jais-3.
 - Ran distributed training experiments on a **SLURM** cluster nodes of **H100** GPUs.
 - Designed and implemented the Arabic custom reward functions for Math and Code experiments.
 - Collaborated and effectively communicated with the post-training team to lift Math and Code performance from **39% to 46%** after rounds of SFT and RL with **GRPO**.
- **R&D Machine Learning Engineer at Yalla Group Limited** Dubai, UAE
Full-time May 2025 - July 2025
 - Developed an AI agent to play the classical board game Draw Dominoes, leveraging deep reinforcement learning.
 - Implemented the agent using the Neural Fictitious Self-Play (NFSP) algorithm, achieving a 72.5% win rate against a uniform random baseline—1.5% higher than the previous benchmark.
 - Designed and deployed a distributed Actor-Learner architecture for scalable training using **Ray**, enabling 24 parallel actors for experience generation and centralized gradient aggregation at the learner.
 - The whole system was deployed on Google Kubernetes Engine (GKE) on GCP.
- **Natural Language Processing (NLP) Engineer Intern at SpeedLegal** Remotely, California, USA
Full-time Jul 2022 - Jun 2024
 - Collaborated in a team to build a Minimum Viable Product in 2 months to automate contract review, helping lawyers extract salient information efficiently.
 - Developed Retrieval-Augmented Generation **RAG** pipelines using LLMs (**GPT-4**), and evaluated IR performance using precision@k, recall@k, and MRR, achieving a high recall@6 of 96% for NDA agreements.
 - Fine-tuned a pre-trained BERT model for extractive question answering on license agreements, achieving 20% exact match and 65% clause-level accuracy, reducing contract review time by 15 minutes.
 - Explored model interpretability using **LIME** to explain classification decisions on legal text, improving model transparency and reliability.

SKILLS SUMMARY

- **Languages and Frameworks:**
C, Python, MongoDB, Unix scripting, Numpy, Pandas, Scikit-learn, PyTorch, Redis, Docker, Ray, Google Cloud Platform (GCP), GitLab CI, Streamlit, Design Patterns, Version Control, RESTful API, SOLID principles, Object Oriented Programming, Operating Systems, Agile Software Development, Reinforcement learning, Information retrieval, Distributed Training, Kubernetes, CI/CD, Model Deployment & Monitoring, Experiment tracking (Weights & Biases / MLflow), LangChain, Agentic Frameworks, Agents, Transformers, Evaluation frameworks.
- **Languages:**
Arabic: Native, English: Fluent

PUBLICATIONS

- **Improving Low-Resource Machine Translation via Round-Trip Reinforcement Learning**
*Ahmed Attia**, Alham Fikri Aji 2026
 - ArXiv Preprint
- **Accurate and Diverse LLM Mathematical Reasoning via Automated PRM-Guided GFlowNets**
*Adam Younsi**, *Ahmed Attia**, Abdalgader Abubaker, Mohamed El Amine Seddik, Hakim Hacid, Salem Lahlou 2025
 - ArXiv Preprint

REFERENCES

Available upon request.